ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P1H Worked Solutions

Jun 2022 | P1H

29 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Sequences**

Jun 2022 P1H - Jun 2022 | P1H

1 Here are the first five terms of an arithmetic sequence.

1 5 9 13 17

(a) Find an expression, in terms of n, for the nth term of this sequence.

....

(2)

The nth term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m, for the $(2m)$th term of this sequence.

....

(1)

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Sequences**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Sequences. The tag is correct.

Solution

The sequence is

$$1,\ 5,\ 9,\ 13,\ 17$$

The common difference is 4, so the nth term is

$$4n - 3$$

For the second sequence, the nth term is $3n + 5$.
The $(2m)$th term is

$$3(2m) + 5 = 6m + 5$$

FINAL ANSWER

$4n - 3$, and $6m + 5$.

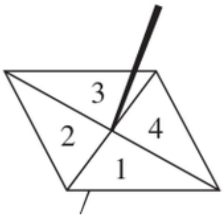
PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

Jun 2022 P1H - Jun 2022 | P1H

2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Probability Toolkit**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

The known probabilities are

$$0.26 + 0.18 = 0.44$$

So the probabilities for numbers 2 and 4 add to

$$1 - 0.44 = 0.56$$

They are equal, so each is

$$0.56 \div 2 = 0.28$$

If the expected frequency for number 3 is 45,

$$0.18n = 45$$

$$n = 250$$

Expected frequency for number 4:

$$250 \times 0.28 = 70$$

FINAL ANSWER

0.28, 250 spins, 70.

PAST PAPER QUESTION Q#: 3 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Jun 2022 P1H - Jun 2022 | P1H

3 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

....
(2)

(b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

....
(2)

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

For the HCF of 56 and 84:

$$56 = 2^3 \times 7$$

$$84 = 2^2 \times 3 \times 7$$

Take the smaller powers:

$$\text{HCF} = 2^2 \times 7 = 28$$

For the LCM of 60 and 72:

$$60 = 2^2 \times 3 \times 5$$

$$72 = 2^3 \times 3^2$$

Take the larger powers:

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

FINAL ANSWER

(a) 28, (b) 360

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2022 P1H - Jun 2022 | P1H

4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

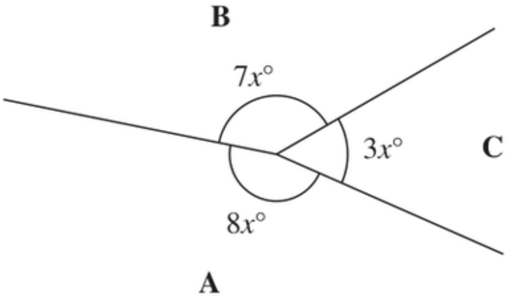

Diagram **NOT** accurately drawn

Polygon **B** has n sides.

Work out the value of n.

$n =$ ....

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Angles in Polygons & Parallel Lines. The tag is correct.

Solution

The angles around the point add to 360°:

$$7x + 3x + 8x = 360$$

$$18x = 360$$

$$x = 20$$

The interior angle of polygon B is

$$7x = 7 \times 20 = 140^\circ$$

So the exterior angle is

$$180^\circ - 140^\circ = 40^\circ$$

For a regular polygon,

$$n = \frac{360}{40} = 9$$

FINAL ANSWER

$n = 9$.

PAST PAPER QUESTION

Q#: 5 Marks: 5

TOPIC: **Solving Linear Equations**

Jun 2022 P1H - Jun 2022 | P1H

5 (a) Expand and simplify $(n-6)(n+4)$

....
(2)

(b) Solve $2x - 3 = \dfrac{3x - 5}{4}$

Show clear algebraic working.

$x =$ ....
(3)

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 5

TOPIC: **Solving Linear Equations**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Corrected to solving linear equations. The original tag was expanding brackets, but the equation is the larger part.

Solution

(a)

$$(n-6)(n+4) = n^2 - 2n - 24$$

(b)

$$2x - 3 = \frac{3x-5}{4}$$

$$8x - 12 = 3x - 5$$

$$5x = 7$$

FINAL ANSWER

(a) $n^2 - 2n - 24$, (b) $x = \frac{7}{5}$

PAST PAPER QUESTION Q#: 6 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Jun 2022 P1H - Jun 2022 | P1H

6 Asha bought an apartment.

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	**Service charge band**
$x \geqslant 700\,000$	A
$600\,000 \leqslant x < 700\,000$	B
$500\,000 \leqslant x < 600\,000$	C
$400\,000 \leqslant x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

(a) Has the annual service charge band changed for Asha's apartment?
Show your working clearly.

(3)

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

(b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

.... %

(3)

(Total for Question 6 is 6 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 6

TOPIC: **Compound Interest & Depreciation**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Compound Interest and Depreciation. The tag is correct.

Method

For part (a), the value in 2021 was 4% more than in 2020.

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

In 2020, $600000 < x < 700000$, so the service charge band was B.
In 2021, 634400 is also between 600000 and 700000, so the band was still B.
For part (b), a depreciation of 15% has multiplier 0.85.
After 2 years, the multiplier is

$$0.85^2 = 0.7225$$

So the value left is 72.25%, meaning the total depreciation is

$$100\% - 72.25\% = 27.75\%$$

FINAL ANSWER
(a) No, it stayed in band B. (b) 27.75%

PAST PAPER QUESTION

Q#: 7 Marks: 4

TOPIC: **Standard & Compound Units**

Jun 2022 P1H - Jun 2022 | P1H

7 A cylinder is placed on the ground.

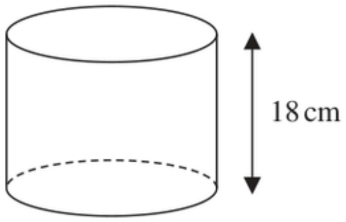

Diagram **NOT** accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.
The pressure on the ground due to the cylinder is 1.4 newtons/cm^2

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

Work out the volume of the cylinder.
Give your answer correct to 3 significant figures.

.... cm^3

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Standard & Compound Units**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Correct. This is a pressure and cylinder volume question.

Solution

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$1.4 = \frac{72}{\text{area}}$$

$$\text{area} = \frac{72}{1.4} = 51.428\ldots\ \text{cm}^2$$

The base area of the cylinder is $51.428\ldots\ \text{cm}^2$.
Volume is

$$51.428\ldots \times 18 = 925.714\ldots$$

FINAL ANSWER
$926\ \text{cm}^3$ to 3 significant figures.

PAST PAPER QUESTION

Q#: 8 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2022 P1H - Jun 2022 | P1H

8 (a) Write 0.000 089 in standard form.

....

(1)

(b) Write 8.34×10^4 as an ordinary number.

....

(1)

(Total for Question 8 is 2 marks)

PAST PAPER SOLUTION Q#: 8 Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$0.000089 = 8.9 \times 10^{-5}$$

$$8.34 \times 10^{4} = 83400$$

FINAL ANSWER
(a) 8.9×10^{-5}, (b) 83400

PAST PAPER QUESTION Q#: 9 Marks: 4

Topic: **Algebraic Roots & Indices**

Jun 2022 P1H - Jun 2022 | P1H

9 (a) Simplify $8 \times (4t)^0$

....

(1)

$x^6 \div x^{-5} = x^p$

(b) Find the value of p

$p =$ ....

(1)

(c) Simplify fully $(2k^2m^4)^3$

....

(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Algebraic Roots & Indices**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Algebraic Roots and Indices. The tag is correct.

Method

$$8 \times (4t)^0 = 8 \times 1 = 8$$

For part (b),

$$x^6 \div x^{-5} = x^{6-(-5)} = x^{11}$$

So

$$p = 11$$

For part (c),

$$(2k^2m^4)^3 = 2^3k^{2\times3}m^{4\times3}$$

$$= 8k^6m^{12}$$

FINAL ANSWER

(a) 8, (b) $p = 11$, (c) $8k^6m^{12}$.

PAST PAPER QUESTION

Q#: 10 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P1H - Jun 2022 | P1H

10 Two circles, C_1 and C_2, are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates (–1, 3) and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates (7, 18) and the radius of C_2 is 6 cm.

(a) Work out the distance between the centre of C_1 and the centre of C_2

.... cm

(3)

(b) Explain why circle C_1 intersects circle C_2

....

(1)

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Coordinate geometry. The tag is correct.

Solution

The centres are

$$C_1 = (-1, 3), \qquad C_2 = (7, 18)$$

The distance between the centres is

$$\sqrt{(7-(-1))^2 + (18-3)^2}$$

$$= \sqrt{8^2 + 15^2} = \sqrt{289} = 17$$

The radii are 3 cm and 15 cm.

$$3 + 15 = 18$$

Since the distance between the centres is 17 cm, which is less than the sum of the radii, the circles intersect.

FINAL ANSWER

17 cm, and the circles intersect because $17 < 18$.

PAST PAPER QUESTION

Q#: 11 **Marks: 5**

TOPIC: **Factorising**

Jun 2022 P1H - Jun 2022 | P1H

11 (a) Factorise $9x^2 - 4y^2$

....

(2)

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

....

(3)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 5

TOPIC: **Factorising**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

(a)

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2 = (3x - 2y)(3x + 2y)$$

(b)

$$\frac{7}{8} - \frac{x+3}{4x} = \frac{7x}{8x} - \frac{2(x+3)}{8x}$$

$$= \frac{7x - 2x - 6}{8x} = \frac{5x - 6}{8x}$$

FINAL ANSWER

(a) $(3x - 2y)(3x + 2y)$, (b) $\frac{5x-6}{8x}$

PAST PAPER QUESTION Q#: 12 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

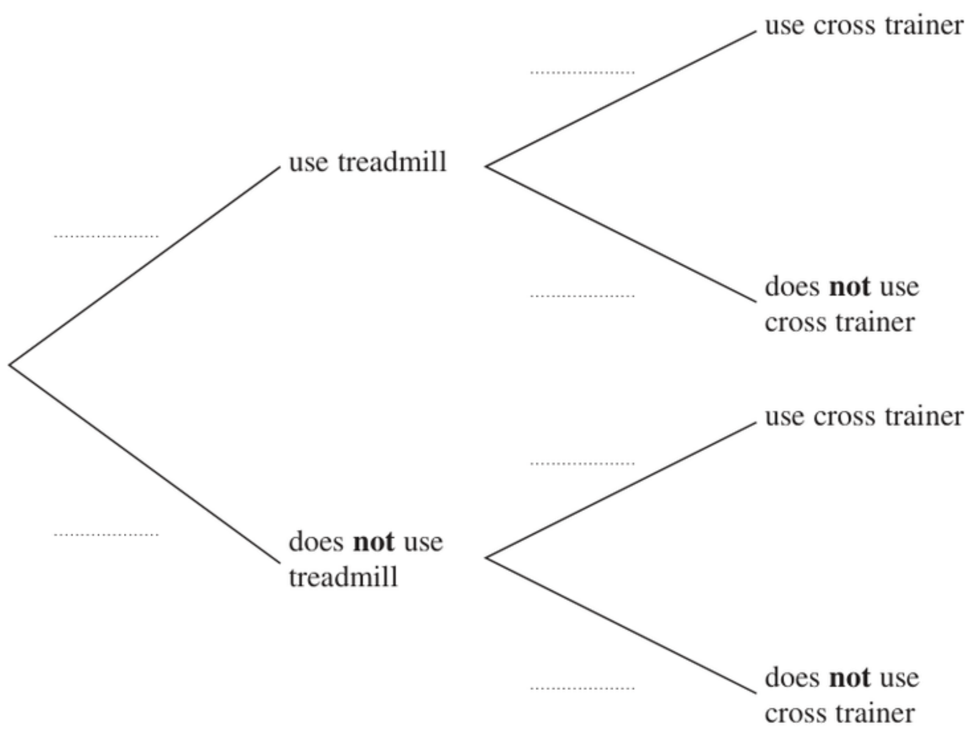
Jun 2022 P1H - Jun 2022 | P1H

12 Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3
When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.



(2)

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

....

(2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

The probability Rudolph uses both machines is

$$0.8 \times 0.3 = 0.24$$

FINAL ANSWER

0.24.

PAST PAPER QUESTION Q#: 13 Marks: 5

TOPIC: **Percentages**

Jun 2022 P1H - Jun 2022 | P1H

13 Antoine is going on holiday.
He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$406

Work out how much Antoine has to pay for Payment 2

\$ ....

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 5

TOPIC: **Percentages**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

Payment 1 is

$$\frac{3}{8}$$

Payment 2 is

$$45\% = \frac{45}{100} = \frac{9}{20}$$

Together:

$$\frac{3}{8} + \frac{9}{20} = \frac{15}{40} + \frac{18}{40} = \frac{33}{40}$$

So Payment 3 is

$$1 - \frac{33}{40} = \frac{7}{40}$$

Payment 3 is \$406, so

$$\frac{7}{40} \times \text{total} = 406$$

$$\text{total} = 406 \times \frac{40}{7} = 2320$$

Payment 2 is

$$45\% \times 2320 = 1044$$

FINAL ANSWER

\$1044

PAST PAPER QUESTION

TOPIC: **Functions**

Jun 2022 P1H - Jun 2022 | P1H

Q#: 14 Marks: 4

14 The function f is defined as

$$f : x \mapsto \frac{2x}{x-6} \qquad x \neq 6$$

(a) Find f(10)

....

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \ldots$

$f^{-1} : x \mapsto$ ....

(3)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

Topic: **Functions**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Functions. The tag is correct.

Solution

$$f(x) = \frac{2x}{x-6}$$

(a)

$$f(10) = \frac{2(10)}{10-6} = \frac{20}{4} = 5$$

(b) Let

$$y = \frac{2x}{x-6}$$

$$y(x-6) = 2x$$

$$yx - 6y = 2x$$

$$x(y-2) = 6y$$

$$x = \frac{6y}{y-2}$$

So

$$f^{-1}(x) = \frac{6x}{x-2}$$

FINAL ANSWER

$f(10) = 5$, and $f^{-1}(x) = \dfrac{6x}{x-2}$.

PAST PAPER QUESTION Q#: 15 Marks: 4

TOPIC: **Probability Toolkit**

Jun 2022 P1H - Jun 2022 | P1H

15 Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.
When Abraham draws a game, he scores 0 points.
When Abraham loses a game, he scores -5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 4

TOPIC: **Probability Toolkit**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

To have a total score of 0 after three games:

- all three games are draws, or
- one game is won and two games are lost.

$$P(\text{DDD}) = 0.5^3 = 0.125$$

$$P(\text{one win and two losses}) = 3(0.3)(0.2)^2 = 0.036$$

$$0.125 + 0.036 = 0.161$$

FINAL ANSWER

0.161.

PAST PAPER QUESTION Q#: 16 Marks: 3

TOPIC: **Surds**

Jun 2022 P1H - Jun 2022 | P1H

16 Without using a calculator, show that $\dfrac{12}{\sqrt{2}-1} - \left(\sqrt{2}\right)^5 = 2\sqrt{32} + 12$

Show your working clearly.

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Surds**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

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Topic check. Corrected to Surds.

Proof

Rationalise the denominator:

$$\frac{12}{\sqrt{2}-1} = \frac{12(\sqrt{2}+1)}{(\sqrt{2}-1)(\sqrt{2}+1)}$$

$$= 12(\sqrt{2}+1) = 12\sqrt{2}+12$$

Also,

$$(\sqrt{2})^5 = (\sqrt{2})^4\sqrt{2} = 4\sqrt{2}$$

So the left side is

$$12\sqrt{2}+12-4\sqrt{2} = 8\sqrt{2}+12$$

And

$$2\sqrt{32}+12 = 2(4\sqrt{2})+12 = 8\sqrt{2}+12$$

So both sides are equal.

FINAL ANSWER

Shown.

PAST PAPER QUESTION **Q#: 17** **Marks: 5**

TOPIC: **Differentiation**

Jun 2022 P1H - Jun 2022 | P1H

17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geqslant 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geqslant 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

.... m

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 5

TOPIC: **Differentiation**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Correct. This is differentiation applied to motion.

Solution

$$s = 4t^2 + \frac{125}{t}$$

Velocity is

$$v = \frac{ds}{dt} = 8t - \frac{125}{t^2}$$

When $v = 0$,

$$8t - \frac{125}{t^2} = 0$$

$$8t^3 = 125$$

$$t^3 = 15.625$$

$$t = 2.5$$

Now find s:

$$s = 4(2.5)^2 + \frac{125}{2.5}$$

$$s = 25 + 50 = 75$$

FINAL ANSWER

75 m.

PAST PAPER QUESTION

Q#: 18 **Marks: 5**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2022 P1H - Jun 2022 | P1H

18 Here is triangle *ABC*

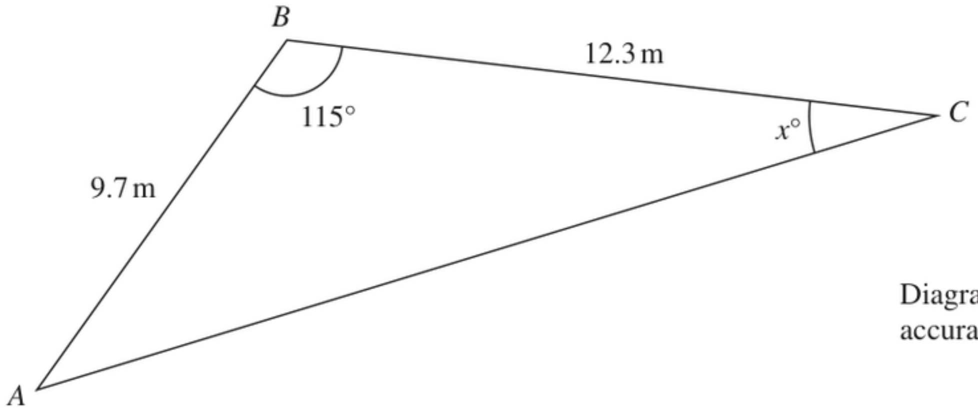

Diagram **NOT** accurately drawn

Work out the value of x

Give your answer correct to 3 significant figures.

$x =$ ....

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION

Q#: 18 **Marks: 5**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2022 P1H - Jun 2022 | P1H

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Topic check. Corrected to Sine, Cosine Rule & Area of Triangles. The question needs the cosine rule and sine rule.

Solution

First use the cosine rule to find AC:

$$AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos 115°$$

$$AC = 18.607\ldots$$

Now use the sine rule:

$$\frac{\sin x}{9.7} = \frac{\sin 115°}{18.607\ldots}$$

$$\sin x = \frac{9.7 \sin 115°}{18.607\ldots}$$

$$x = 28.194\ldots°$$

FINAL ANSWER

28.2°.

PAST PAPER QUESTION Q#: 19 Marks: 5

Topic: **Circles, Arcs & Sectors**

Jun 2022 P1H - Jun 2022 | P1H

19

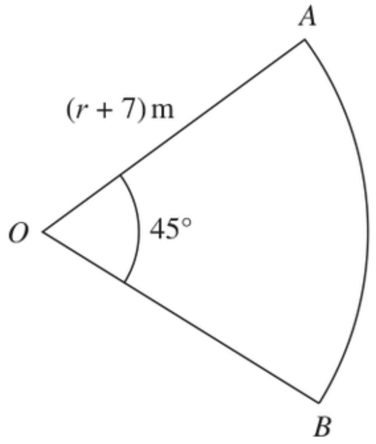

Diagram **NOT** accurately drawn

OAB is a sector **S** of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45°$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r
Show your working clearly.

Question 19 continued.

$r =$ ....

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

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WORKED SOLUTION

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Topic check. Circles, Arcs & Sectors. The tag is correct.

Solution

Area of sector S:

$$\frac{45}{360}\pi(r+7)^2 = \frac{1}{8}\pi(r+7)^2$$

Area of circle C:

$$\pi(r-2)^2$$

The sector area is twice the circle area:

$$\frac{1}{8}\pi(r+7)^2 = 2\pi(r-2)^2$$

$$(r+7)^2 = 16(r-2)^2$$

Since $r > 2$,

$$r + 7 = 4(r-2)$$

$$r + 7 = 4r - 8$$

$$r = 5$$

FINAL ANSWER

$r = 5$.

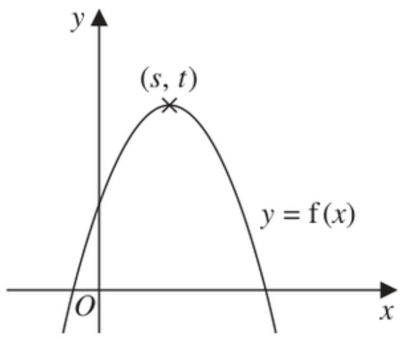
PAST PAPER QUESTION

Q#: 20 **Marks: 2**

TOPIC: **Transformations of Graphs**

Jun 2022 P1H - Jun 2022 | P1H

20 The diagram shows a sketch of part of the curve with equation $y = \mathrm{f}(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (s, t)

Find, in terms of s and t, the coordinates of the maximum point on the curve with equation

(i) $y = \mathrm{f}(x - 2)$

(.......................... , ...)
(1)

(ii) $y = 3\mathrm{f}(x)$

(.......................... , ...)
(1)

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The maximum point of $y = f(x)$ is (s, t).
For

$$y = f(x - 2)$$

the graph moves 2 units right:

$$(s, t) \to (s + 2, t)$$

For

$$y = 3f(x)$$

the y-coordinate is multiplied by 3:

$$(s, t) \to (s, 3t)$$

FINAL ANSWER
$(s + 2, t)$ and $(s, 3t)$

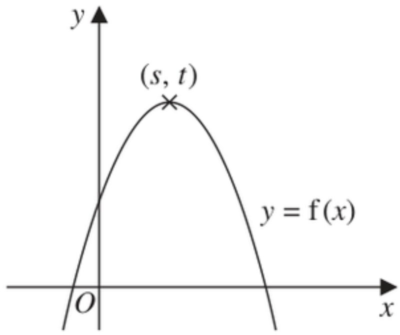
PAST PAPER QUESTION

Q#: 20 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2022 P1H - Jun 2022 | P1H

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (s, t)

Find, in terms of s and t, the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

(.......................... , ...)
(1)

(ii) $y = 3f(x)$

(.......................... , ...)
(1)

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The maximum point of $y = f(x)$ is (s, t).
For

$$y = f(x-2)$$

the graph moves 2 units right:

$$(s,t) \to (s+2, t)$$

For

$$y = 3f(x)$$

the y-coordinate is multiplied by 3:

$$(s,t) \to (s, 3t)$$

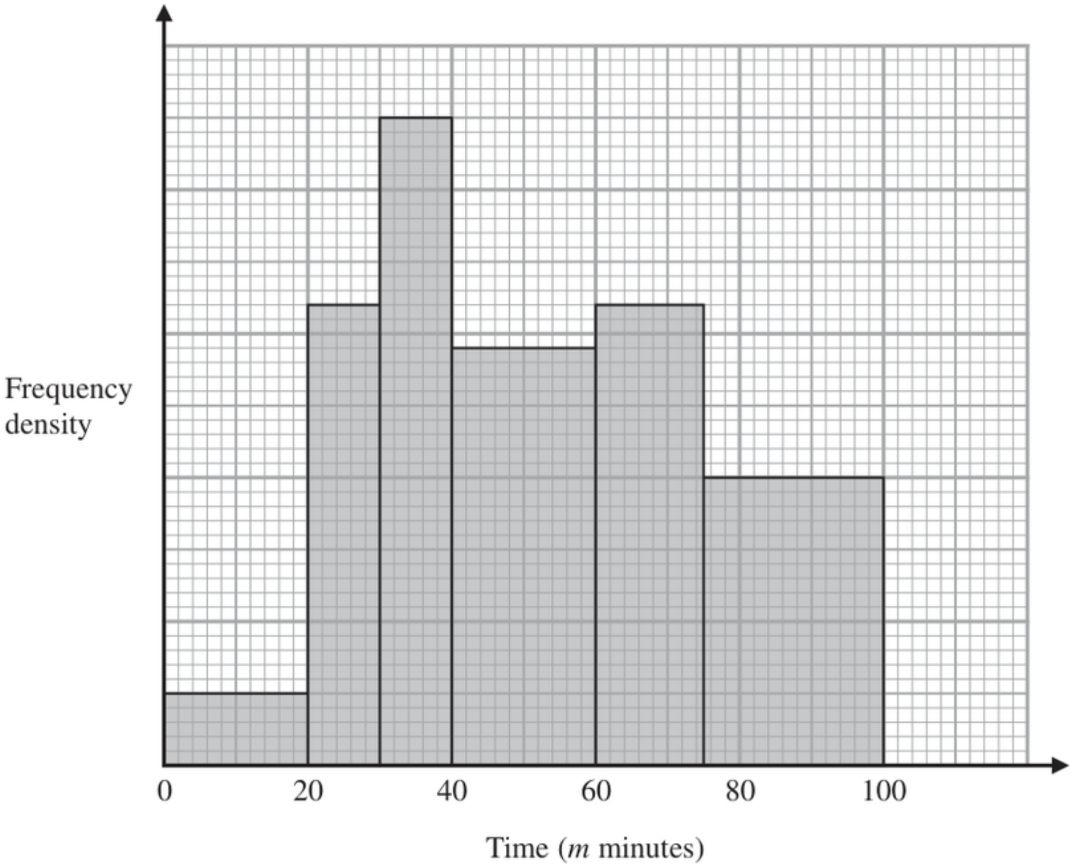
FINAL ANSWER
$(s+2, t)$ and $(s, 3t)$

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Histograms**

Jun 2022 P1H - Jun 2022 | P1H

21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$

Work out an estimate for the number of children for whom $50 < m \leqslant 80$

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Histograms**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $m \leq 20$, the frequency is 10 and the class width is 20, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

Using this scale from the histogram:

$$40 < m \leq 60 \quad \text{has frequency density } 2.9$$

$$60 < m \leq 75 \quad \text{has frequency density } 3.2$$

$$75 < m \leq 100 \quad \text{has frequency density } 2.0$$

For $50 < m \leq 80$, use part of each relevant bar:

$$(10)(2.9) + (15)(3.2) + (5)(2.0)$$

$$= 29 + 48 + 10$$

$$= 87$$

FINAL ANSWER

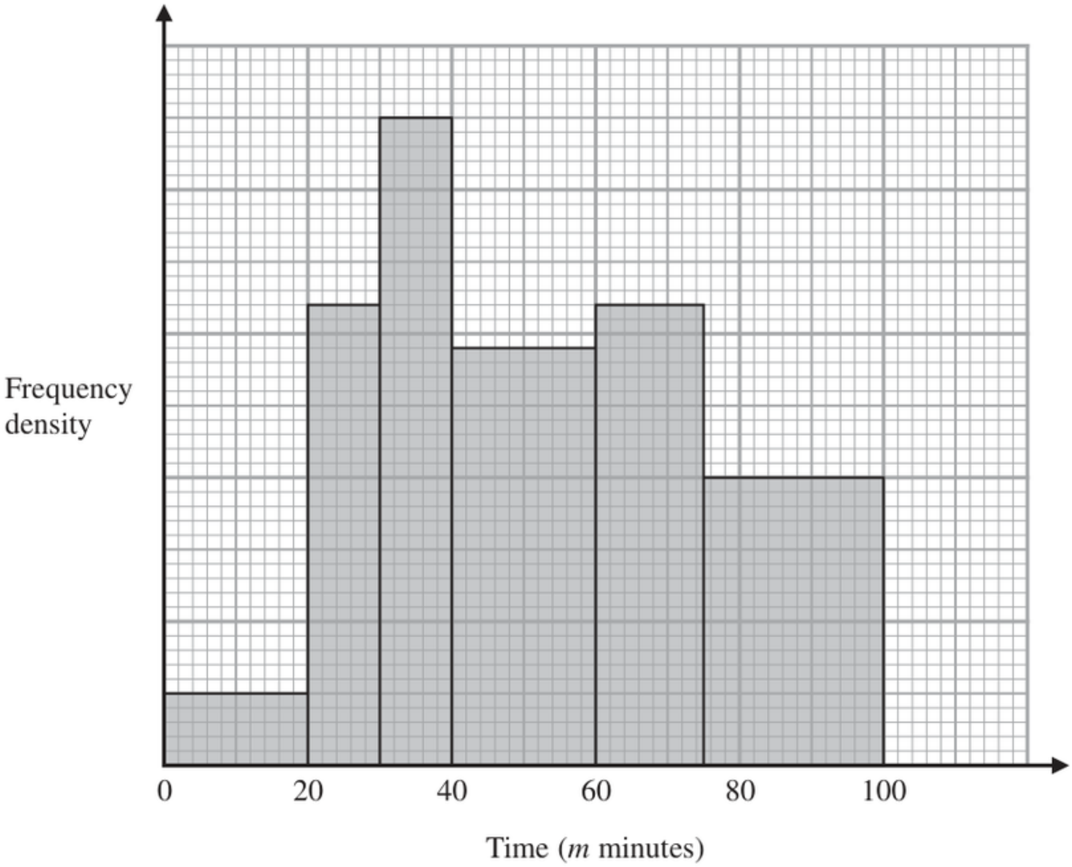
87 children

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Histograms**

Jun 2022 P1H - Jun 2022 | P1H

21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$

Work out an estimate for the number of children for whom $50 < m \leqslant 80$

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 3

TOPIC: **Histograms**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

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$$= 29 + 48 + 10$$

$$= 87$$

FINAL ANSWER

87 children

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2022 P1H - Jun 2022 | P1H

22 A solid is made from a cone and a hemisphere.

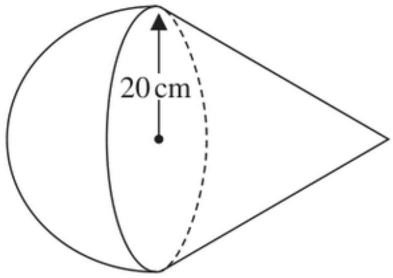


Diagram **NOT** accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi \text{ cm}^2$

The volume of the solid is $k\pi \text{ cm}^3$

Work out the exact value of k

$k =$ ....

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION **Q#: 22** **Marks: 5**

TOPIC: **Volume & Surface Area**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The cone has radius 20 cm.
The curved surface area of the cone is

$$\pi r l$$

So

$$\pi(20)l = 580\pi$$

$$l = 29$$

Use Pythagoras to find the height of the cone:

$$h^2 + 20^2 = 29^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

The volume is $k\pi$, so

$$k = \frac{24400}{3}$$

FINAL ANSWER

$\boxed{\frac{24400}{3}}$

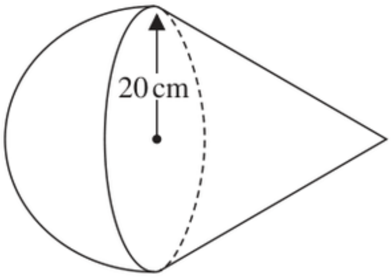
PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2022 P1H - Jun 2022 | P1H

22 A solid is made from a cone and a hemisphere.



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(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Volume & Surface Area**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

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Total volume:

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The volume is $k\pi$, so

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FINAL ANSWER

$$\boxed{\frac{24400}{3}}$$

PAST PAPER QUESTION Q#: 23 Marks: 6

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2022 P1H - Jun 2022 | P1H

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

$$177 \qquad 175 \qquad 173 \qquad 171 \qquad 169$$

Find the value of n
Show clear algebraic working.

Question 23 continued

$n =$ ....

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The interior angles form an arithmetic sequence:

$$177,\ 175,\ 173,\ 171,\ 169, \ldots$$

So

$$a = 177, \quad d = -2$$

For an n-sided polygon, the sum of the interior angles is

$$180(n-2)$$

The sum of the arithmetic sequence is

$$\frac{n}{2}\left(2(177) + (n-1)(-2)\right)$$

$$= \frac{n}{2}(354 - 2n + 2)$$

$$= \frac{n}{2}(356 - 2n)$$

$$= n(178 - n)$$

Set this equal to the polygon angle sum:

$$n(178 - n) = 180(n-2)$$

$$178n - n^2 = 180n - 360$$

$$n^2 + 2n - 360 = 0$$

Factorise:

$$(n+20)(n-18) = 0$$

Since $n > 5$,

$$n = 18$$

FINAL ANSWER

$n = 18$

PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2022 P1H - Jun 2022 | P1H

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Here are the first five terms of this sequence.

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Find the value of n
Show clear algebraic working.

Question 23 continued

$n =$ ....

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

Topic: **Angles in Polygons & Parallel Lines**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

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Method

The interior angles form an arithmetic sequence:

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$$n^2 + 2n - 360 = 0$$

Factorise:

$$(n+20)(n-18) = 0$$

Since $n > 5$,

$$n = 18$$

FINAL ANSWER

$n = 18$

PAST PAPER QUESTION

Q#: 24 Marks: 4

Topic: **Completing the Square**

Jun 2022 P1H - Jun 2022 | P1H

24 Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Completing the Square**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$q + 12x - qx^2 = -q\left(x^2 - \frac{12}{q}x\right) + q$$

Complete the square:

$$x^2 - \frac{12}{q}x = \left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}$$

So

$$q + 12x - qx^2 = -q\left[\left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}\right] + q$$
$$= q + \frac{36}{q} - q\left(x - \frac{6}{q}\right)^2$$

Compare with

$$a - b(x - c)^2$$

FINAL ANSWER

$a = q + \dfrac{36}{q}$, $b = q$, $c = \dfrac{6}{q}$

PAST PAPER QUESTION

Q#: 24 **Marks: 4**

TOPIC: **Completing the Square**

Jun 2022 P1H - Jun 2022 | P1H

24 Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 24 is 4 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 4

TOPIC: **Completing the Square**

Jun 2022 P1H - Jun 2022 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$q + 12x - qx^2 = -q\left(x^2 - \frac{12}{q}x\right) + q$$

Complete the square:

$$x^2 - \frac{12}{q}x = \left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}$$

So

$$q + 12x - qx^2 = -q\left[\left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}\right] + q$$

$$= q + \frac{36}{q} - q\left(x - \frac{6}{q}\right)^2$$

Compare with

$$a - b(x - c)^2$$

FINAL ANSWER

$a = q + \frac{36}{q}$, $b = q$, $c = \frac{6}{q}$